Simulation and reconstruction chain of the inclusive and coherent cos 2φ measurements with a tensor-polarized ^{6}Li beam at the EIC: assumptions, methodology and closure

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2026-08-24 · rewritten 2026-08-28 after a full review of the toolchain (Appendix A) · companion to Report 1 (the projections) and Report 3 (the machine and the detector) · computed with `polligen.reco`, `polligen.hfs` and `polligen.recopseudo` on the PYTHIA 8 hadronic final state of `tools/pythia8`; every number is reproducible with the commands of `docs/reproduction_manual.md`

Abstract. The double-helicity-flip structure function $\Delta(x,Q^2)$ of ^{6}Li is projected in Report 1 as the cos 2φ modulation of inclusive DIS on a transversely tensor-polarized beam, and as a modulation of the recoil azimuth in coherent diffraction with the ^{6}Li detected intact. This report describes the simulation and reconstruction chain behind those projections — the polarized event kernel, the PYTHIA 8 hadronic final state, the electron-side, hadron-side and far-forward detector responses, the kinematic reconstruction, the spin-state-sorted estimator and the conversion to Δ — lists every assumption it rests on with its provenance and leverage, and reports its closure at the reconstructed level. The analysis path consumes only quantities an experiment records — the spin state of each bunch crossing, the tensor polarization and luminosity of each state, the reconstructed electron, the hadronic $E - p_z$ sum and the Roman-Pot angle pair — with simulation-derived corrections applied as functions of reconstructed variables; a 2026-08-28 review of the toolchain found one place where that was not so (the hadronic-scale calibration, now keyed on the reconstructed point), a truncation of the PYTHIA library below $x = 16/s$ (now regenerated), and two statistical defects of the estimator (now fixed), none of which changes a headline number by more than its statistical error. At e(10 GeV) × ^{6}Li(99.5 GeV/u) the four sweet-spot bins reconstruct with purities 0.56–0.75 and amplitude errors $0.9–3.0 \times 10^{-4}$ in one year, the fit closing on the response-folded truth; the coherent channel closes at the lithium tagging optics of Report 1 with $2.5–6.0 \times 10^6$ tagged recoils per year, unbiased in every $|t|$ bin with more than 10^5 recoils.

1 · Scope

Report 1 presents the measurement; Report 3 the accelerator and detector it is projected on. This report is the layer between them: what the simulation generates, what it pretends the detector records, what the analysis is allowed to read, and how well the chain returns what was put in. Section 2 walks the chain stage by stage with the boundary between generator truth and measured quantities made explicit; Section 3 collects the assumptions; Section 4 gives the methodology of each step; Section 5 reports the reconstructed-level closure; Section 6 the systematic effects that were measured on the chain and the specifications they imply; Section 7 what the chain does not model. The audit of the code that produced this revision is summarized in Appendix A and recorded in `docs/code_review_2026-08-28.md`.

2 · The chain

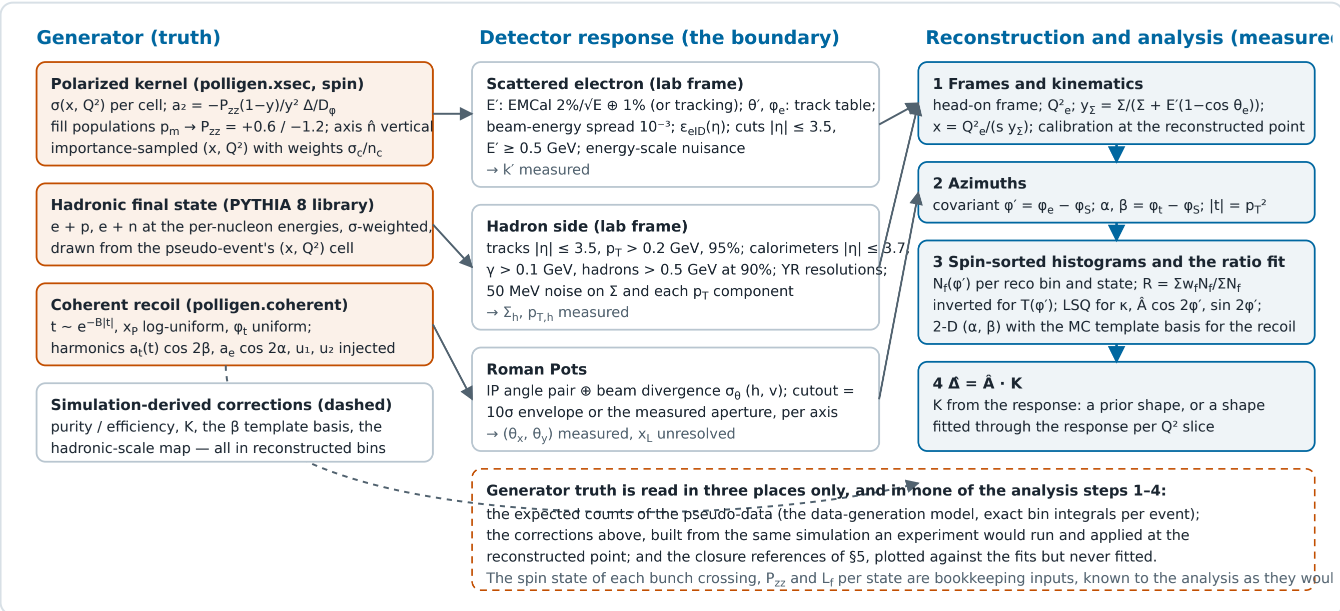
The observable is a Fourier coefficient of a *ratio* of spin-sorted azimuthal distributions. For an unpolarized electron on a spin-1 target with alignment axis at polar angle θ_S and azimuth φ_S about the virtual photon, the Hoodbhoy–Jaffe–Manohar cross section of the projection state m is [1,2]

$$\frac{d\sigma^{(m)}}{dx\,dy\,d\phi} = \frac{2y\alpha^2}{Q^2}\left[F_1 + \tfrac{2}{3}a_m b_1 + \frac{1-y}{xy^2}\left(F_2 + \tfrac{2}{3}a_m b_2\right) - \frac{1-y}{y^2}\,c_m \sin^2\theta_S\,\Delta(x,Q^2)\cos 2\phi\right], \qquad c_m = 3m^2 - 2,$$

so that a fill with populations p_m along a transverse axis has the azimuthal distribution $1 + a_2 \cos 2\varphi'$, $\varphi' = \varphi - \varphi_S$, with

$$a_2 = -P_{zz}\frac{1-y}{y^2}\frac{\Delta}{D_\phi}, \qquad D_\phi = F_1 + \frac{1-y}{xy^2}F_2, \qquad \hat{A} \equiv \frac{a_2}{P_{zz}}, \qquad P_{zz} = \sum_m p_m c_m.$$

The chain that turns this into a measurement has six stages (Schematic 1, Table 1). **(i) The polarized kernel.** `polligen.xsec` evaluates the master formula per (x, Q^2) cell for each projection state m of the fill, with the spin-density matrix of `polligen.spin` providing the m-populations of the two fill types and the covariant azimuth of §4.1 providing φ′; the unpolarized cross section, F_2 and $R = \sigma_L/\sigma_T$ come from `polli_fastsim.structure` and the Δ model from `polli_fastsim.delta_models`. **(ii) The hadronic final state.** The tensor polarization enters only the azimuthal weight of the inclusive cross section, so the hadronic system of each pseudo-event is taken from an unpolarized PYTHIA 8 library of e + p and e + n events at the same beam energies (§4.2), weighted by their cross sections. **(iii) The detector response.** The scattered electron is smeared in the detector frame with the calorimeter, tracking and electron-identification models of §3; the hadronic final state passes through a hadron-side response — tracker and calorimeter coverage, thresholds, efficiencies, Yellow-Report resolutions, calorimeter noise — in the same frame; the coherent recoil is measured by an emulated Roman-Pot system that returns the angle pair at the interaction point smeared by the beam divergence and cut by the near-beam envelope. **(iv) Reconstruction.** The head-on frame is restored, Q^2 comes from the electron, y from the Σ method on the hadronic $E - p_z$ sum, x from the two (the mixed method), and the azimuths from the covariant formula on the reconstructed four-vectors. **(v) The estimator.** Events are histogrammed in φ′ per spin state and per reconstructed (x, Q^2) bin, and the cos 2φ′ coefficient is fitted to the spin-state ratio, which cancels any azimuthal acceptance common to the two states. **(vi) Extraction.** The amplitude is converted to Δ with a bin-centering factor derived from the response — either from an assumed Δ shape, or from a shape fitted through the response itself.



Schematic 1 — the chain, and the boundary between truth and measurement. Left: what is generated. Middle: the three detector responses, which consume the truth and emit measured quantities. Right: the analysis, which reads measured quantities, the bookkeeping of the run plan, and corrections derived from the simulation in reconstructed bins.

STAGE	INCLUSIVE E ⁶LI → E′ X	COHERENT E ⁶LI → E′ X ⁶LI(G.S.)	TRUTH READ	CODE
Generation	importance-sampled (x, Q^2) over a 60×45 log grid with the generator window loosened (y ≥ 0.004, Q^2 ≥ 0.7 GeV^2, |η| ≤ 3.8, E′ ≥ 0.3 GeV) so events can migrate into the analysis window; 400 pseudo-events per cell, weight σ_c/n_c	|t| exponential with B = 50 GeV^{-2}, $x_P \in [10^{-3}, 10^{-2}]$, φ_t uniform; 6×10⁶ recoils; injected harmonics of §4.5	—	`recopseudo.RecoResponse`, `CoherentResponse`
Hadronic final state	PYTHIA 8.311 e+p / e+n at the per-nucleon	not generated (no electron side; α	the pseudo-event's (x, Q^2,	`hfs.HFSLibrary`, `HFSResponse`; `tools/pythia8`

	energies, $Q^2 \geq 0.7$ GeV², $\hat{m} \geq 0.5$ GeV, lepton radiation off; 8 M events over the three configurations; a library event from the pseudo-event's (x, Q^2) cell supplies the captured fraction of Σ_h and the p_T ratio	integrated analytically)	y), to draw and scale the library event	
Detector response	e′: lab-frame smearing of E', θ', φ_e, beam-energy spread, $\varepsilon_{eID}(\eta)$; hadrons: coverage, thresholds, efficiencies, resolutions, noise, in the lab frame	Roman Pots: angle pair ⊕ divergence, rectangular cutout (envelope or aperture per axis)	the true four-vectors, to smear them	`reco.smear_electron`, `hfs.HadronResponse`, `reco.rp_measure`
Reconstruction	head-on frame; $Q^2{}_e$; y_Σ (JB, DA available); $x = Q^2{}_e/(s\, y_\Sigma)$; hadronic-scale calibration at $(x_{mixed}, Q^2{}_e)$; analysis cuts $Q^2 \geq 1$, $0.01 \leq y \leq 0.95$, $W^2 \geq 10$, $\lvert\eta\rvert \leq 3.5$, $E' \geq 0.5$; covariant φ'	$\lvert t\rvert = p_T{}^2$, $\beta = \varphi_t - \varphi_S$ from the smeared angle pair; $\alpha = \varphi_e - \varphi_S$	none	`reco.electron_method`, `hfs.hadronic_kinematics`, `reco.azimuth_wrt_lepton_plane`
Pseudo-data	exact expected counts per spin state and φ' (or (α, β)) bin, from the per-event amplitude at the true kinematics and the reconstructed bin assignment, with the φ'-efficiency harmonic (common or per fill) and the relative-luminosity offset switched on; Poisson-fluctuated at 10 and 100 fb⁻¹/u		the per-event a_2 and dilution (data generation)	`RecoResponse.expected_counts`, `CoherentResponse.expected_counts_2d`
Estimator	spin-state ratio per bin, inverted for $T(\varphi')$, weighted LSQ for κ, $\hat{A}$, $\hat{A}_{sin}$	the same in (α, β) with the acceptance-weighted β template basis from the response, the u_1, u_2 nuisances subtracted, and three parity-forbidden sine columns as null tests	the basis and the truth references only	`reco.harmonic_ratio_fit`, `harmonic_ratio_fit_2d`
Extraction	$\hat{\Delta} = \hat{A}\, K$, $K = \Delta_{prior}(x_c, Q^2{}_c)/\mathrm{fold}(\Delta_{prior})$ or from the shape fitted through the response	$a_t(t_{ref})$, a_e per $\lvert t\rvert$ bin	the response (fold, purity, efficiency)	`recopseudo.fold`, `fold_shape_fit`, `delta_from_amplitude`

Table 1 — the chain stage by stage, and where generator truth is read. The pseudo-data are generated from the truth, as data are generated by nature; the reconstruction and the estimator read measured quantities; the corrections are what an experiment takes from its own simulation.

3 · Assumptions

Table 2 lists every input the chain consumes, with its value, its provenance and what a wrong value would move. Three classes are distinguished: inputs the accelerator or the detector will fix (to be asked for precisely; Report 3 carries their full specification), physics-model inputs (scenarios, bracketed by bands in Report 1), and idealisations of this analysis (to be replaced or quantified as the chain matures). Idealisations that are structural — no radiative corrections, no backgrounds, no coherent electron side — are in §7.

INPUT	VALUE USED	PROVENANCE AND STATUS	LEVERAGE IF WRONG
Beam and spin			
^{6}Li energies	`40.8 / 99.5 / 137.5 GeV/u`	derived: γ-matched to the 41 / 100 / 275 GeV proton, rigidity-capped at the top [7,13]; the chain publishes at the mid configuration	every far-forward cut (∝ A p_u); the sweet-spot y values
beam divergence σ_θ (h / v)	`220/380, 180/180, 92/92 µrad`	YR Table 10.1 proton values, gamma-matched species step [7] (`reco.sigma_theta_for`); no lithium optics is published	coherent tag acceptance, exponentially; Δp/p 7–10 × 10^{-4} enters only the pot dispersion term
tagging optics (coherent channel)	β^*_x `× 50 / 176 / 89, pots following the 10σ envelope 0.33 × 3.8, 0.17 × 1.8, 0.12 × 0.92 mrad`	this programme's proposal (Report 1 §6.1): the optimum of acceptance × luminosity with the vertical plane at high acceptance, L/L_{HA} = 1/7.1, 1/13.3, 1/9.5; assumes parallel-to-point transport and pots that move	without it no recoil is tagged at any configuration (Figure 2c)
P_{zz} of the two spin states	`+0.6 / −1.2, equal luminosity`	in-ring placeholder (ANL source target ≥ 0.8); the m = 0-rich state at $-2P_{zz}$ assumes the same source purity; taken as exactly known by the analysis	statistical error ∝ $1/P_{zz}$; a scale error is a 1:1 normalisation on Δ (§6)
spin axis	`vertical,` $\varphi_S = \pi/2$, $\theta_S = 90°$	the stable spin direction of the arcs; an azimuthal error is self-calibrated by the sin 2φ′ term (tan 2δ recovered exactly in a test), a polar tilt dilutes by $\sin^2\theta_S$	0.8% dilution at 5° tilt; none from the azimuth
spin-state pattern	`bunch-by-bunch alternation`	required, not assumed: the ratio cancels only an acceptance common to both states (§4.3)	a 10^{-3} difference of the cos 2φ′ acceptance harmonic between the states fakes 5.6 × 10^{-4}
luminosity	`10 / 100` fb^{-1} `per nucleon`	YR convention for one / ten years [7]; shares 0.5/0.5 with a 10^{-3} offset unknown to the analysis	statistics; the offset enters the ratio as a constant
Central detector			
EMCal σ_E/E for e′	`2%/√E ⊕ 1%`	YR requirement of the backward crystals [7]; correct where the sweet-spot electrons sit (η = −2.9 to −1.6); the η-dependent table changes $\hat{\Delta}$ by ≤ 0.1%	$\delta Q^2/Q^2$ (subdominant to the angles)
tracking σ_p/p	`√((a p)² + b²), η-piecewise`	YR requirement (Fig. 8.3 / Table 11.2), endcap constants padded; a requirement, not an achievement	only with `energy="best"`
track angular resolution	`5 / 3 / 2 / 1 mrad by η (3 mrad at the sweet spots)`	**placeholder** — no YR angular requirement exists; it is also the only stand-in for the 81–211 µrad beam-divergence floor on θ′	$\delta Q^2/Q^2$ = 5.2% at spots 1–2; a factor ≲ 3 bracket (purity 0.66 → 0.70 at a third of it)
electron identification	$\varepsilon_{eID}(\eta)$ `= 0.70–0.95`	constructed between the ATHENA and ECCE proposal values; no ePIC curve; no purity or background model	none on $\hat{A}$ beyond statistics — a fill-independent efficiency cancels in the ratio (an η tilt of 0.05 moves $\hat{\Delta}$ by 0.02%)
electron energy scale	`nuisance ±1%`	unknown to the analysis; d ln x/d ln E′ = 2 − y for the Σ method	0.3–1.0% on $\hat{\Delta}$ per 1%
hadron-side coverage and thresholds	`tracks |η| ≤ 3.5,` p_T `> 0.2 GeV, 95%; calorimeters |η| ≤ 3.7; γ > 0.1 GeV; hadrons > 0.5 GeV, 90%`	stand-ins of the right magnitude in the detector frame (the ePIC nominal forward reach is 4.0; B0 at η = 4.6–5.9 not modelled)	the captured fraction of Σ_h (0.70–0.85 at the sweet spots, Table 3); 4.0 recovers a quarter of the forward loss with δy/y unchanged

hadron-side resolutions	YR tables per region	requirement values [7], with the YR 2% EMCal constant	subdominant to the noise below y = 0.02
calorimeter noise on Σ_h	50 MeV Gaussian per event on Σ and each p_T component	a one-parameter stand-in for the noise and threshold floor whose leverage at low y Arratia et al. show for H1/ATHENA [18]; no ePIC number exists (plans/04 #21)	the y ≈ 0.01 resolution: 0.22 → 0.54 → 1.01 at y = 0.005 for 0 → 50 → 100 MeV
hadronic energy scale	calibrated from the simulation at (x_{mixed}, Q^2_e); residual ±2% as a nuisance	the analysis's own calibration; keyed on reconstructed variables (2026-08-28)	0.1–0.6% on Â per 1% residual
Far forward			
Roman-Pot measurement	angle pair ⊕ σ_θ (h, v); no detector term	beam effects dominate the ePIC far-forward resolution [15]; x_L unresolved; the dispersion of the off-momentum recoil (x_P ≤ 0.01, below every divergence) is folded into the horizontal term of the tagging optics	|t| smearing $2p_T\delta p_T$; the φ_t dilution the template basis absorbs
pot aperture	2.00/3.00, 1.35/3.00, 1.03/2.30 mrad (h/v)	measured with an intact ^{6}Li in a September-2024 ePIC geometry (`tools/fullsim`); the pots have moved since	with fixed pots the tag is 10^{-8}–10^{-26}; the tagging optics assumes pots that follow
exclusivity	pure coherent sample	no breakup background, vetoes or Z-identification modelled (plans/06); the incoherent yield is m-state blind and dilutes rather than fakes	the |t|-fit purity of Report 1 §6.4
Physics model			
F_2, F_1, R	toy F_2 (±37% of CT18 over the acceptance); $R = 0.18/(1 + Q^2/50)$	placeholders with PDF-grid and R1998 backends behind the same hooks (Report 1 §3.2); F_2 cancels exactly in the amplitude, R does not	rates and statistical errors to ×1.5; Δ/F_1 by +18 / +19 / +8 / −4% at the four spots if R1998 replaces the placeholder
$\Delta(x, Q^2)$	moment-constrained ansatz A, dilution 1/3; α_s from the CT18NLO table	Report 1 §4; the LO α_s fallback (used without the `parton` package, now announced) raises the normalisation by 14%	the injected amplitudes, not the closure
coherent scenario	$f_0 = 0.04$, $B = 50$ GeV^{-2}, $\varepsilon_{B0} = -0.08$, $a_e = 0.01$, $u_1 = 0.05$, $u_2 = 0.02$	Report 1 §6; u_1, u_2 at the ZEUS 1σ edge [13,14], taken as known by the fit	rates and the injected harmonics; δu_2 leaks into a_e at first order (§6)
$a_t(t)$ template	linear in |t|	the anchor's low-|t| behaviour [9]; one coefficient per |t| bin	a different shape is a model systematic to be varied
radiative corrections	none in the published response; collinear ISR available as a hook (`polligen/radiative.py`, `RecoResponse(isr=…)`, default off)	bounded, not calculated: the exponentiated leading-log spectrum [21,22] as a kinematic migration. Collinear ISR fakes no φ′ modulation and cancels in the spin-state ratio; $x = Q^2_e/(s\, y_\Sigma)$ is exact and only the Q^2_e label migrates, by $1/(1 - z)$. Tensor-sector RC uncalculated (plans/05 §5.5), and no unpolarized study bounds it	Δ by +0.5 to +1.2% at the four sweet spots (mean over eight response seeds; ≤2.9% once the generator window is opened below Q^2 = 0.15 GeV² and the low-Q^2 feed-in saturates); ≤0.25% with a HERA-style $E - p_z$ window. §7

Table 2 — the assumptions. "Requirement" means the Yellow Report value, which is a design target and not a measured performance. Every number in this report is conditional on this table.

4 · Methodology

4.1 · Frames and azimuths

The generator works in the head-on frame (ion along +z, electron along −z); the detector sees the lab frame, in which the electron beam defines the ePIC axis and the ion enters at −25 mrad in x. The two are related by a rotation of 12.5 mrad about y and a boost β_x = sin 12.5 mrad; energies change by 8×10^{-5} and the vertical spin direction not at all. Every detector effect — the electron smearing, the hadron acceptance and thresholds, the Roman-Pot angles — is applied in the lab frame, and the measured four-vectors are transformed back before any kinematic quantity is formed, as an ePIC analysis does. For the electron the transformation changes only odd harmonics of the azimuth (≈ 10^{-3} in cos φ, cos 2φ untouched to 10^{-17}), so it cannot fake the observable; for the hadrons it moves the forward calorimeter edge by up to ±0.25 units of η around the ring, which is worth +0.7 to +1.5% of the captured Σ_h at the sweet spots relative to a head-on-frame acceptance.

The azimuth of the observable is the covariant one used for every transverse-spin azimuth in SIDIS and exclusive analyses [3,4], applied to the spin four-vector S and, for the recoil, to P′:

$$\cos\phi_S = -\frac{l_\perp \cdot S_\perp}{|l_\perp||S_\perp|}, \qquad \sin\phi_S = -\frac{\varepsilon_{\mu\nu\rho\sigma} l^\mu S^\nu P^\rho q^\sigma}{(P\cdot q)\sqrt{1+\gamma^2}\,|l_\perp||S_\perp|}, \qquad v_\perp^\mu = g_\perp^{\mu\nu} v_\nu, \quad \gamma^2 = \frac{M^2 Q^2}{(P\cdot q)^2}.$$

With the axis transverse at lab azimuth φ_S this reduces exactly to $\varphi' = \varphi_e - \varphi_S$ for a massless target and to $O(\gamma^2)$ for the nucleus — below 5 mrad at the sweet spots of the mid and top configurations ($\gamma^2 \le 0.02$) and 12 mrad at those of the low configuration — and it is verified against an explicit collinear-frame construction to 10^{-13}. cos 2φ′ is even under φ′ → −φ′ and φ′ → φ′ + π and the axis is headless, so no orientation convention enters; a misalignment δ of the assumed axis turns the modulation into cos 2δ cos 2φ′ + sin 2δ sin 2φ′, and the fitted sin 2φ′ term returns tan 2δ — the self-calibration of the axis. The tensor structure functions b_1–b_4 leak into cos 2φ′ at $O(\gamma^2)$ with a bound of 1.15 $\gamma^2\, b_1/F_1$ (the full Cosyn et al. combination [12]), ≤ 0.15% of everything reported here and subtractable with the A_{zz} of the same data.

4.2 · Kinematic reconstruction and the hadronic final state

The electron method gives Q^2 well ($\delta Q^2/Q^2 = \delta E'/E' \oplus \cot(\theta'/2)\,\delta\theta'$, 5.2% at the two Q^2 = 1.14 GeV² spots with the 3 mrad angular placeholder, 2.9% and 1.6% at the other two) but loses y as (1 − y)/y: three of the four sweet spots sit at y = 0.010–0.025 (Table 3), where a 1.2% energy resolution gives δy/y = 0.46–1.2 and the bins are not reconstructible. The remedy is HERA's [5,6]: y from the hadronic final state,

$$y_\Sigma = \frac{\Sigma}{\Sigma + E'(1-\cos\theta_e)}, \qquad \Sigma = \sum_{h\neq e'} (E_h - p_{z,h}), \qquad x = \frac{Q_e^2}{s\, y_\Sigma},$$

with θ_e from the ion direction. Σ is insensitive to particles lost along the ion beam (each contributes $E - p_z = m_T e^{-y}$) and y_Σ to collinear initial-state radiation; the mixed x inherits both, because the factor (1 − z) of a radiated photon cancels between Q^2_e and s — it is the Q^2 label of the bin that migrates under radiation, not x (plans/08 D3 corrected; the migration is bounded in §7). The Jacquet–Blondel and double-angle methods are implemented alongside and checked against hand-derived formulas; the Σ method is the best of the three below y = 0.05 and is used throughout.

The hadronic final state is a PYTHIA 8.311 production of 8 M unpolarized e + p and e + n events at the three per-nucleon beam energies ($Q^2 \ge 0.7$ GeV², lepton radiation off, $\hat{m} \ge 0.5$ GeV so that the library covers the whole x range of the generator window — its 4 GeV default had silently removed x < 16/s, 39% of the selected rate, before 2026-08-28), merged in the ratio of their cross sections. It enters as a library: for a pseudo-event at (x, Q^2) a library event of the same cell, passed through the hadron-side response with its noise off, supplies the captured fraction of Σ_h and the p_T ratio, which are applied to the pseudo-event's true $\Sigma_h = 2E_e y + m_N^2/(E_N + p_N)$ — the massless value plus the same target-mass term the library's Σ carries, 4.4 MeV at 99.5 GeV/u — and the 50 MeV noise is added per pseudo-event; y_Σ then follows with the pseudo-event's own reconstructed electron. Table 3 and Figures 3–4 give the result at the mid configuration: 80–92% of the true Σ_h is within the acceptance and above threshold, 7–17% escapes forward beyond |η| = 3.7 (the target-fragmentation side of a W = 6–10 GeV system; the lithium fragments at η ≈ 8 never enter, each carrying only $m^2/2p$), 1–6% lies below threshold, and 70–85% is captured through the full response. The loss is a scale bias on y_Σ, calibrated as an analysis would calibrate it: the library's own events, smeared and reconstructed like the pseudo-events, give a map of $\langle\Sigma_{reco}\rangle/\langle\Sigma_{true}\rangle$ in bins of the reconstructed (x_{mixed}, Q^2_e), and each pseudo-event's sums are divided by the map at its own uncalibrated reconstructed point. (Until 2026-08-28 the map was looked up at the event's *true* cell, which no experiment can do; the measurable version gives purities within 0.06 of it and is the only one used now.) What sets the resolution is not the loss but the noise: with 50 MeV the Σ method gives $\delta y/y$ = 0.32 / 0.22 / 0.29 / 0.15 at the four spots against 0.19–0.22 without noise, and at y = 0.005 it runs 0.22 → 0.33 → 0.54 → 1.01 for 0 → 25 → 50 → 100 MeV. Each configuration selects its own four bins; at the low and top configurations they resolve to 0.39 / 0.23 / 0.24 / 0.11 and 0.23 / 0.19 / 0.21 / 0.18, so the (0.141, 14.3 GeV²) bin is best measured at 5 × 40.8 and the low-x bins improve with √s.

SWEET SPOT (X, Q²)	Y	W [GEV]	E′: E′, Θ′, Η	ΔY/Y E′ ALONE	Γ_H AS Η · Σ-WEIGHTED PARTICLE Η (16 / 50 / 84%)	WITHIN ACCEPTANCE, |Η| ≤ 3.7 (TRACKS / Γ / HCAL)	LOST FORWARD	BELOW THRESHOLD	CAPTURED THROUGH THE RESPONSE (MEAN · MEDIAN)	ΔY/Y: Σ, JB, DA
0.028, 1.14	0.010	6.3	9.98 GeV, 107 mrad, −2.93	1.18	1.6 · 1.8 / 2.6 / 3.8	0.80 (0.45 / 0.25 / 0.10)	0.17	0.03	0.70 · 0.73	0.32, 0.32, 0.51
0.011, 1.14	0.025	10.0	9.90 GeV, 107 mrad, −2.92	0.46	0.7 · 0.8 / 1.6 / 2.9	0.87 (0.51 / 0.26 / 0.10)	0.08	0.06	0.74 · 0.78	0.22, 0.22, 0.40
0.071, 3.13	0.011	6.5	10.02 GeV, 177 mrad, −2.42	1.06	2.1 · 1.9 / 2.6 / 3.7	0.83 (0.47 / 0.25 / 0.10)	0.15	0.02	0.74 · 0.77	0.29, 0.29, 0.44
0.141, 14.3	0.025	9.4	10.10 GeV, 378 mrad, −1.65	0.45	2.0 · 1.7 / 2.2 / 2.9	0.92 (0.55 / 0.27 / 0.09)	0.07	0.01	0.85 · 0.87	0.14, 0.15, 0.21

Table 3 — the four sweet spots of Report 1 at e(10) × ⁶Li(99.5) and their hadronic final state (PYTHIA 8, 4 M e + p and e + n events; `hfs_acceptance.py`, `hfs_resolution.py`). The acceptance columns are geometry and thresholds only, in the detector frame; "captured through the response" includes efficiencies, the track turn-on and the mass hypotheses, noise off, and equals the median y_Σ/y; the resolutions are 68% half-widths through the full response with 50 MeV of noise. Differences of 0.01 between this table and the figures are Monte-Carlo noise.

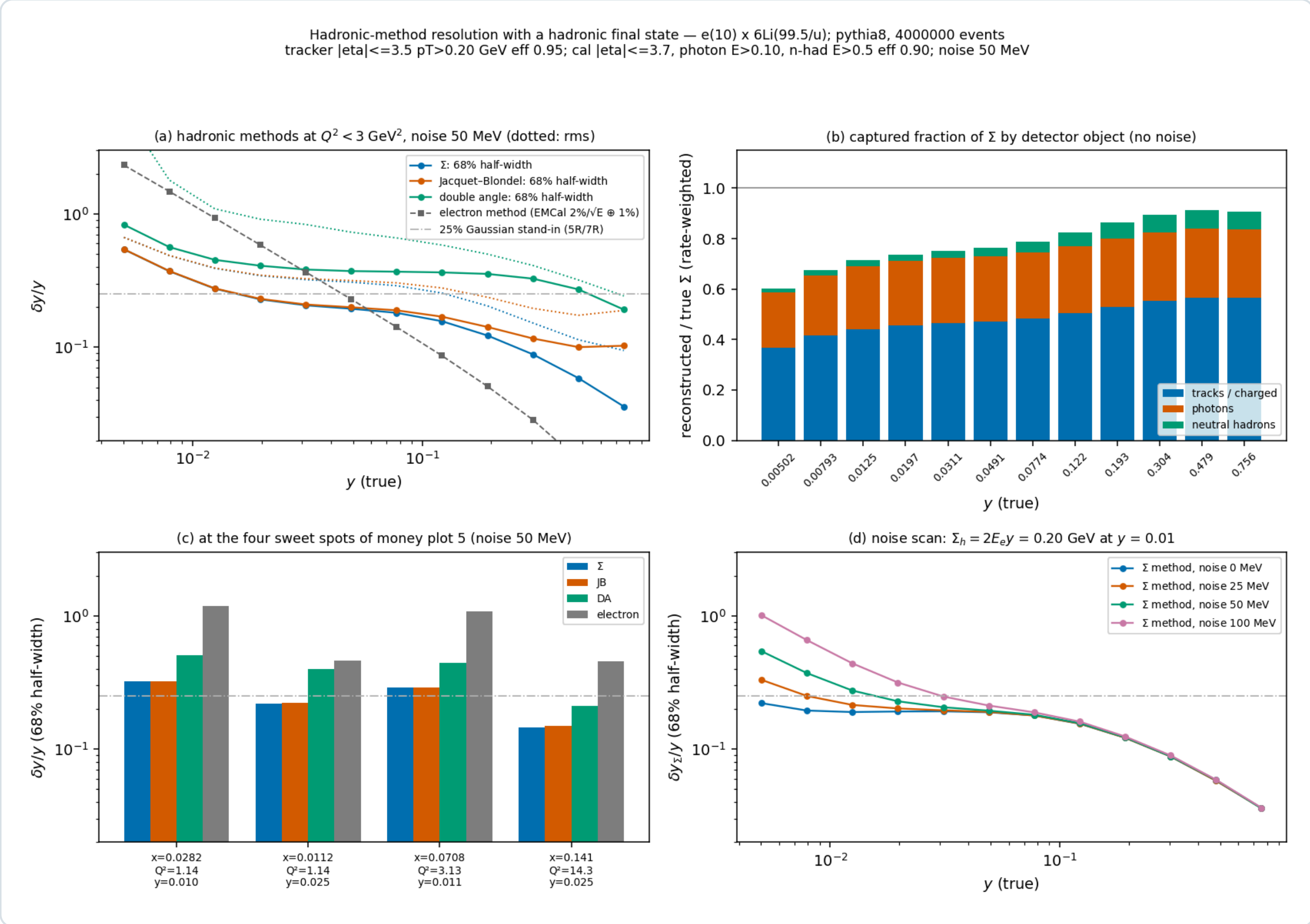


Figure 3 — hadronic-method resolution (e10 × ⁶Li99.5, PYTHIA 8, `hfs_resolution.py`). **(a)** δy/y (68% half-width; dotted: rms) of the Σ, Jacquet–Blondel and double-angle methods at $Q^2 < 3$ GeV² with 50 MeV of noise, against the electron method and the 25% stand-in of the earlier chain. **(b)** Captured fraction of Σ by detector object, rate-weighted, no noise. **(c)** The four sweet spots. **(d)** The Σ-method resolution for 0, 25, 50 and 100 MeV of noise: the floor decides the y ≈ 0.01 bins.

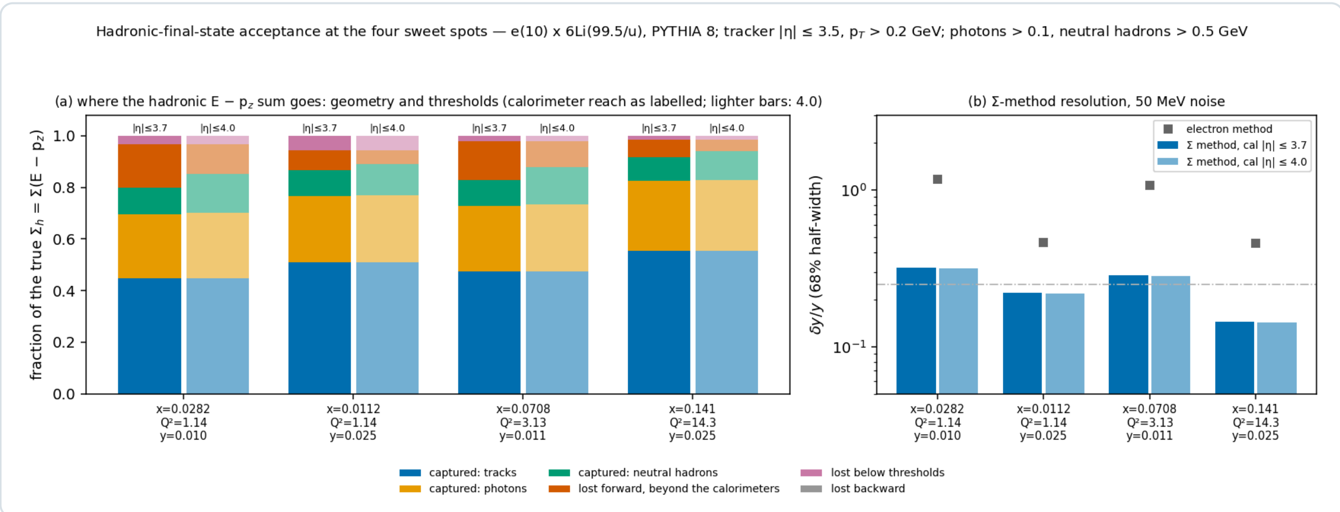


Figure 4 — where the hadronic E − p_z sum goes (e10 × ⁶Li99.5, PYTHIA 8, `hfs_acceptance.py`). **(a)** The true Σ_h by fate — captured as tracks, photons and HCal objects; lost forward beyond the calorimeters, below threshold, or backward — for a calorimeter reach of |η| ≤ 3.7 (full) and the ePIC nominal 4.0 (light). **(b)** The Σ-method δy/y through the full response for both reaches: the extra reach moves the bias, not the resolution.

4.3 · The spin-state-sorted estimator

A single-fill fit of N(φ′) measures the detector's own cos 2φ′ acceptance harmonic divided by P_{zz} on top of the physics: a 3% harmonic aligned with the vertical axis — the symmetry axis of the crossing-angle plane, of module boundaries and of the Roman-Pot cutout — fakes 0.05, four to eleven times the projected amplitudes (Figure 1d). The measurement therefore sorts the events of each reconstructed bin by spin state f, histograms $N_f(\varphi')$, and forms the bin-wise ratio

$$R(\phi')=\frac{\sum_f w_f N_f(\phi')}{\sum_f N_f(\phi')},\qquad w_f=P_f-\bar{P},\qquad \bar{P}=\frac{\sum_f L_f P_f}{\sum_f L_f},\qquad R=\frac{\sigma_P^2\,T(\phi')}{1+\bar{P}\,T(\phi')},\qquad T=\kappa+\hat{A}\cos 2\phi',\qquad \sigma_P^2=\frac{\sum_f L_f(P_f-\bar{P})^2}{\sum_f L_f},$$

which is exact for yields $N_f = L_f\,\varepsilon(\varphi')\,\sigma\,[1 + P_f T]$ because $\Sigma_f L_f w_f = 0$: any efficiency common to the spin states cancels bin by bin, and a relative-luminosity error enters only through $\bar{P}$, as a φ′-independent constant plus a second-order rescaling. R is inverted for T bin by bin (with the exact Jacobian in the error propagation) and T fitted by weighted least squares to $\kappa + \hat{A}\cos 2\varphi' + \hat{A}_s \sin 2\varphi'$, the finite-bin dilution sin w/w divided out; the bin variances use the expected per-state counts so that a bin populated by one state alone keeps a finite weight. Two states of equal luminosity give $\delta\hat{A} = 2\sqrt{(2/N)}/(P_+ - P_0)$: the m = ±1-rich state at $P_+ = +0.6$ alternated with the m = 0-rich state at $P_0 = -1.2$ is 1.5× more precise than a single $P_{zz} = 0.6$ fill with the whole luminosity, and an unpolarized reference state would cost a factor two instead. κ is the transverse tensor rate shift of the b_1 sector, measured as a by-product. The cancellation holds only for an acceptance *common* to the states: a difference $\delta\varepsilon_2$ of the cos 2φ′ harmonic between them enters as $\Sigma_f\, l_f(P_f - \bar{P})\, e_f/\sigma_P^2 = \delta\varepsilon_2/(P_+ - P_0)$ at any luminosity split, so the states must alternate bunch by bunch within a fill, or the harmonic be stable to 10^{-4} between fills (§6). The same estimator in two azimuths serves the coherent channel (§4.5).

R itself is unbiased at any occupancy — conditional on a bin's total the per-state split is multinomial and R is linear in it — but the *inversion* is not. $\hat{T} = R(1 + u)/(\sigma_P^2 - \bar{P}R)$ is strongly curved over the range $|R| \le \max_f|P_f - \bar{P}|$ that R explores when a bin holds a handful of counts (a bin with one count has R = ±0.9 exactly): its second derivative $2(1 + u)\sigma_P^2\bar{P}/(\sigma_P^2 - \bar{P}R)^3$ carries the sign of $\bar{P}$, so the flip plan's $P_0 = -2P_+$ — $\bar{P} = -0.3$ — makes the inversion concave and each bin carry a negative Jensen offset $(1 + u)\bar{P}/(\sigma_P^2 \nu)$, while the data-driven 1/var weights add an opposite-sign term of the same order. The inclusive φ′ bins carry ≥ 10^4 counts and are immune; the sparse coherent |t| bins are not, and there the offset is common to every α at fixed β (the acceptance is flat in α and modulated ×25 in β by the pot cutout), so it is orthogonal to ⟨cos 2α⟩ and lands on a_t and the constant. The alternative is to keep the per-bin efficiency as a free parameter and profile it out. Because ε_b enters as a linear scale the profile is closed-form, $\hat{\varepsilon}_b = n_b/(1 + u_b + \bar{P}T_b)$, and substituting it back leaves *exactly* the conditional multinomial given the bin totals, $p_{f,b} = l_f(1 + u_b + P_f T_b)/(1 + u_b + \bar{P}T_b)$. Profiling as many nuisance parameters as there are bins would ordinarily raise the incidental-parameter objection; it does not here, because the conditional score has zero mean bin by bin at any count — $\Sigma_f p_f(P_f/q_f - \bar{P}/D)$ vanishes identically — so no 1/ν term survives. Newton from T = 0 converges in six iterations under a step-halving guard that keeps every populated bin's density positive — and a step that guard has halved to nothing is not read as convergence: the gradient at the solution is tested against the error bars it would move, so a fit driven onto the boundary is named rather than returned; the covariance is the conditional Fisher information at the observed totals; empty bins contribute nothing and single-state bins contribute a finite n log p term, which removes the infinite-weight pathology structurally rather than by a variance floor. It is the same model, the same basis and the same assumptions as the ratio — in particular the same blindness to a state-dependent acceptance, measured identical to within 2% of the statistical error — and on exact counts it returns the injected coefficients to machine precision with errors that agree with the ratio's to 0.24%, so no error bar in this report moves. It is available throughout as an option, the ratio remains the default, and §5.2 measures what it buys.

4.4 · Response, calibration and unfolding

The inclusive response is importance-sampled: every accepted cell of a 60 × 45 log-log grid receives 400 pseudo-events, log-uniform in the cell, with weight σ_c/n_c (n_c the draws inside the generator acceptance, so the cell rate is conserved; the amplitude is taken at the cell centre, worth ≤ 0.2% at the sweet spots), and each passes through the chain of Schematic 1. For a reconstructed bin the response gives its purity (the fraction of its rate that originates in the true bin), its efficiency (the fraction of the true-bin rate reconstructed in it after ε_{eID}), and D, the rate- and ε_{eID}-weighted, φ′-diluted amplitude of the events landing in it over the amplitude of the true bin. The expected φ′ counts per spin state are exact bin integrals of $\mathrm{den}_f + P_f\,\hat{A}\cos 2\varphi'$ over those events, with the φ′ efficiency and the luminosity shares of the run plan; they are Poisson-fluctuated at 10 and 100 fb⁻¹/u and the fit is compared with the reconstructed-bin truth, which is what an unbiased estimator on this sample must return.

The amplitude is exactly linear in Δ — Δ enters the master formula in one place — so the response is a linear operator on Δ, fold(Δ) = ⟨(∂A/∂Δ) Δ(x_{true}, Q^2_{true})⟩ over the events of the bin, and the conversion $\hat{\Delta}$ = Â K with K = Δ(x_c, Q^2_c)/fold(Δ) is the bin-centering with migration included. Evaluating K with the shape that generated the pseudo-data is circular and reproduces the injected curve by construction; evaluating it with a wrong prior biases $\hat{\Delta}$ by up to 20–24% over the plotted bins of a Q^2 slice, and by (−4.2, +8.0, −5.6, +4.9)% (model B) or (+8.6, +2.5, +7.3, +3.2)% (flat toy) at the four spots. Fitting a two-parameter tilt of the prior *through* the response per Q^2 slice on a bounded grid — the data choose the shape, the box guards the conditioning — reduces the residual to at most 5.7 / 5.7 / 5.8% (model B) and 4.3 / 4.3 / 3.6% (toy) over the plotted bins of the three slices, and to (−1.5, −1.8, −0.3, +0.5)% and (+4.6, +3.0, +1.8, −0.9)% at the spots. The Δ error bar then carries the statistics, the shape-fit error, the response Monte Carlo error (0.1–1.0% per plotted bin) and the spread over the priors the tilt cannot absorb, the last dominating in the best-measured bins.

4.5 · The coherent channel

The intact recoil keeps the beam rigidity (A/Z = 2, $x_P \le 0.01$), so the pots resolve only its transverse displacement: the reconstruction returns the angle pair (θ_x, θ_y) at the interaction point relative to the beam axis — crossing angle and reference orbit drop out by construction — smeared by the bunch's own divergence, and $p_T = A\, p_u|\theta|$, $\varphi_t = \mathrm{atan2}(\theta_y, \theta_x)$, $|t| = p_T^2$ with x_L set to one ($t_{min} = M^2x_P^2/(1 - x_P) = 3 \times 10^{-3}$ GeV² at $x_P = 0.01$ is a known bias once x_P is measured from M_X). A recoil is tagged when it clears, per axis, the larger of the 10σ beam envelope and the pot aperture. At the Yellow Report optics with the measured aperture the tagged fraction is 7×10^{-8}, 6×10^{-27} and 2×10^{-17} at the three configurations (Figure 2c): the coherent channel exists only under the tagging optics of Report 1 §6.1, in which the horizontal β* is raised to the optimum of acceptance × luminosity, the vertical plane stays at high acceptance, and the pots follow the envelope in both planes — a cutout of 0.33 × 3.8 mrad (0.08 × 0.93 GeV) at 5 × 40.8 with σ_θ = 33 / 380 μrad (the horizontal including the beam's momentum spread through the pot dispersion), giving a 41% tag at 1/7.1 of the high-acceptance luminosity. The closure below is at that point.

With the recoil measured there are two physical azimuths besides the spin: $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, and the spin-sorted yield is modelled as

$$N_f(\alpha,\beta) \propto \varepsilon(\alpha,\beta)[1+u_1\cos(\alpha-\beta)+u_2\cos 2(\alpha-\beta)+P_f(\kappa+a_e\cos 2\alpha+a_t(t)\cos 2\beta+a_m\cos(\alpha+\beta))],$$

where a_t is the deformation term of the polarized-deuteron anchor [9] (∝ |t| at low |t|, coefficient $2a_2$ in the anchor's convention, predicted to flip sign for ^{6}Li), a_e the flat double-helicity-flip term that needs the lepton plane, a_m a nuisance, and u_1, u_2 the L–T and T–T′ interference of unpolarized diffractive DIS [13,14]. The two mechanisms are separated by the azimuth they modulate and by their t dependence. The cutout is far from circular, so its acceptance carries a large ⟨cos 2β⟩ of its own — −0.27 at the tagging cutout after the divergence smearing, −0.93 at the measured aperture — and no single-fill fit survives; the spin-state ratio of §4.3 in (α, β) cancels it bin by bin, and its acceptance-profiled likelihood twin does the same without the low-count bias that the sparsest |t| bins carry (Table 5). Because the cutout correlates β with |t| inside a reconstructed |t| bin, the fit uses an acceptance-weighted basis from the response — the in-bin means of cos β, sin β, cos 2β, sin 2β of the true β per reconstructed β bin, and the template ⟨(t/t_{ref}) cos 2β⟩ of the assumed linear shape — so that the coefficients refer to the unsmeared truth and a_t is quoted at t_{ref}, the rate-weighted mean true |t| of the bin; this is one coefficient per bin, the t dependence of a_t coming from the bins together. The three parity-forbidden partners sin 2α, sin 2β and sin(α + β) are fitted alongside as a null test: a spin-axis error δ gives tan 2δ in both azimuths, a pot roll δ_t gives tan $2\delta_t$ in β alone. α is integrated analytically per event; the coherent pseudo-experiment has no electron side, and x_P is sampled but never reconstructed (the M_X method would provide it).

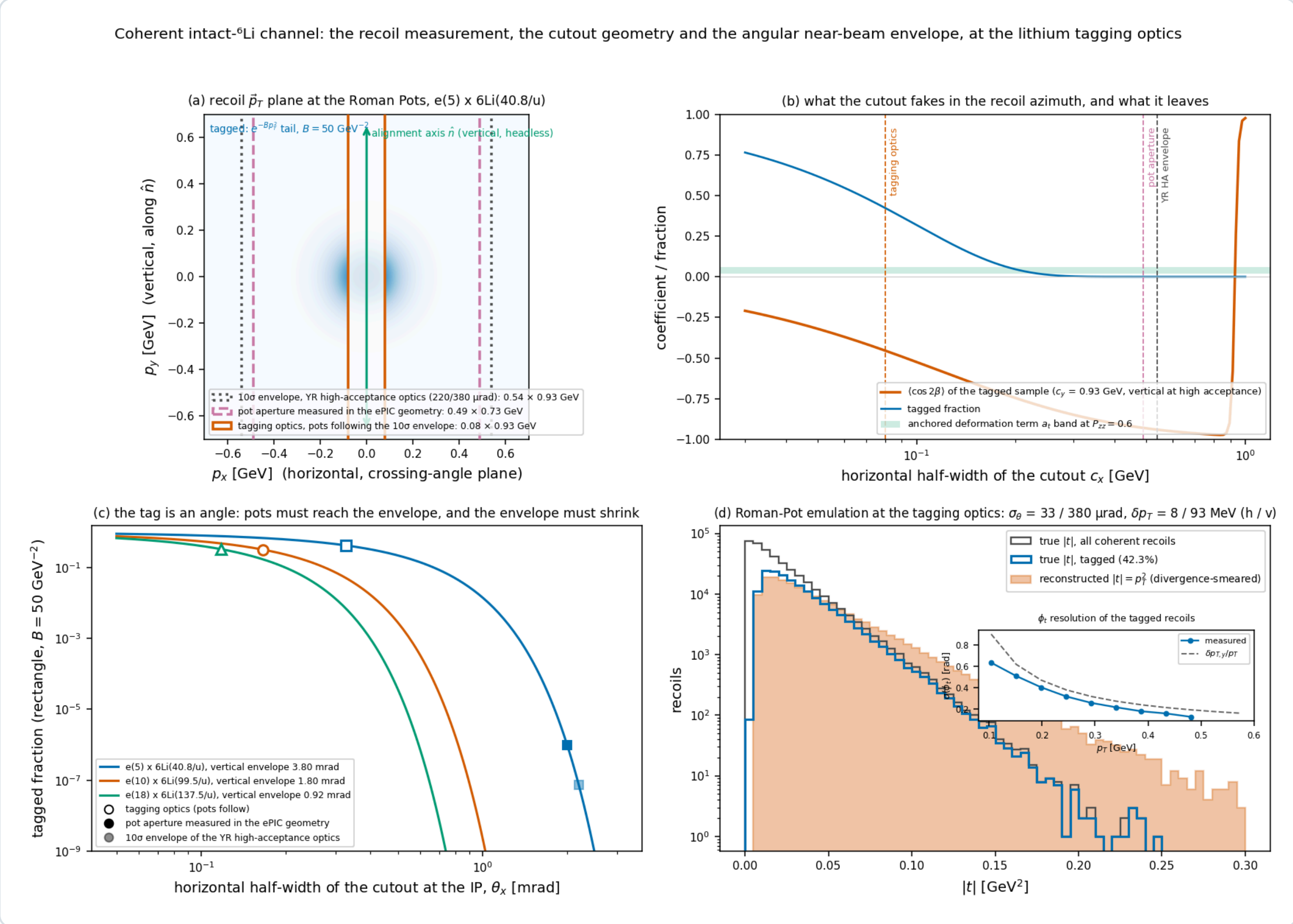


Figure 2 — the coherent measurement quantified (5 × 40.8, `reco_chain_figures.py`). **(a)** The recoil p_T plane with the exponential coherent spectrum and the three cutouts: the Yellow Report high-acceptance 10σ envelope, the pot aperture measured in the ePIC geometry, and the tagging optics with pots following the envelope. **(b)** ⟨cos 2β⟩ of the tagged sample about the vertical spin axis and the tagged fraction versus the horizontal half-width of the cutout, against the anchored deformation band. **(c)** Tagged fraction versus the horizontal half-angle at the interaction point for the three configurations, with the three cutouts marked. **(d)** The Roman-Pot emulation at the tagging optics: true, tagged and reconstructed |t|, and (inset) the φ_t resolution, dominated by the 93 MeV vertical divergence.

4.6 · What the analysis reads

The paths from counts to $\hat{A}$ and to a_t, a_e were traced assignment by assignment in the review (Appendix A). They consume: the spin state of each bunch crossing, the tensor polarization and luminosity share of each state (the run-plan bookkeeping, known to an experiment from the source pattern, the polarimeter and the luminosity monitor); the reconstructed electron four-vector; the hadronic sums and the calibration map at the reconstructed point; the Roman-Pot angle pair after the cut; and the simulation-derived corrections — purity, efficiency, K, the β basis, t_{ref} — in reconstructed bins. Generator truth is read to generate the pseudo-data, to build those corrections, and to plot the closure references; the analysis assumes the nuisances u_1, u_2 and the luminosity shares (as fit arguments, so that a wrong assumption can be exercised) and takes P_{zz} as exactly known (its scale is a normalisation, §6). The bins shown in the figures are selected on their expected count and statistical error alone.

5 · Closure at the reconstructed level

5.1 · Inclusive: money plots 5R and 7R

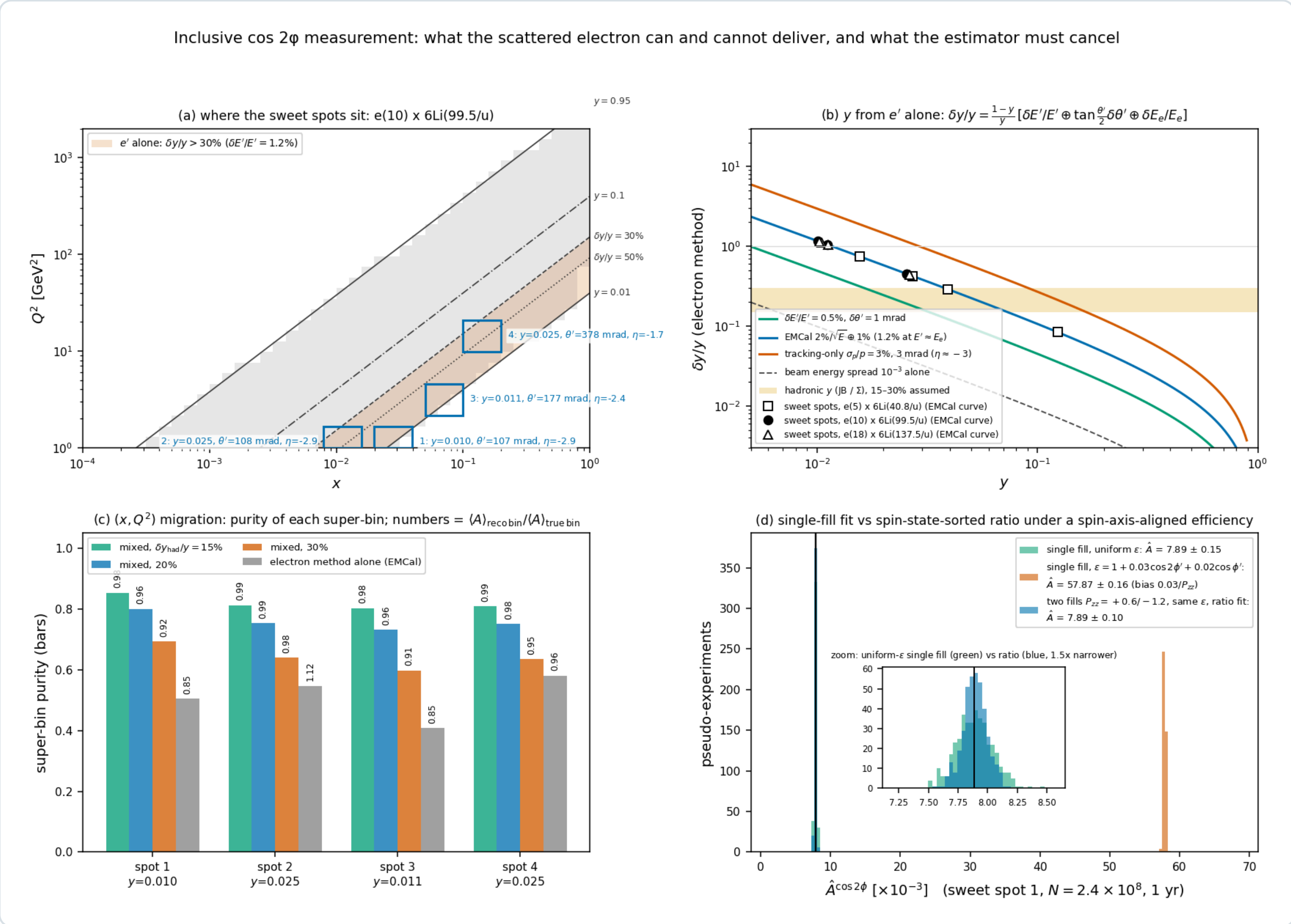
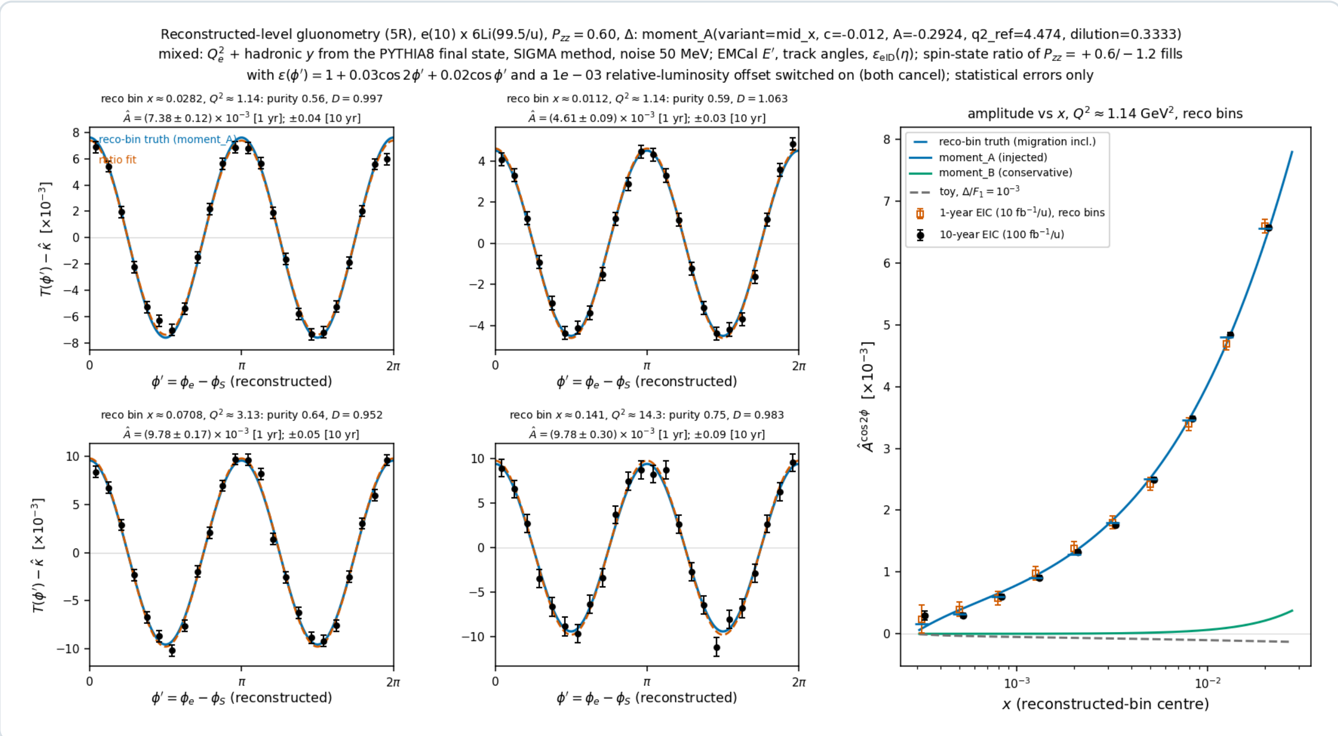


Figure 1 — the inclusive measurement quantified (e10 × ^{6}Li99.5, `reco_chain_figures.py`). **(a)** The analysis acceptance with iso-y lines and the four sweet-spot super-bins; the shaded band is where the electron alone gives δy/y > 30% at a 1.2% energy resolution. **(b)** The electron-method δy/y versus y for calorimeter, tracking and beam-energy-spread inputs, with the sweet spots of the three configurations and the 15–30% hadronic band. **(c)** Purity of each super-bin for three Gaussian hadronic-y resolutions and for the electron method alone; the numbers are the reconstructed-bin over true-bin amplitude. **(d)** Four hundred pseudo-experiments of the spot-1 super-bin at one year: the single-fill fit under a uniform efficiency (green), the same under ε = 1 + 0.03 cos 2φ′ + 0.02 cos φ′ (vermillion, biased by $0.03/P_{zz}$), and the two-state ratio under that efficiency (blue, unbiased and 1.5× narrower).

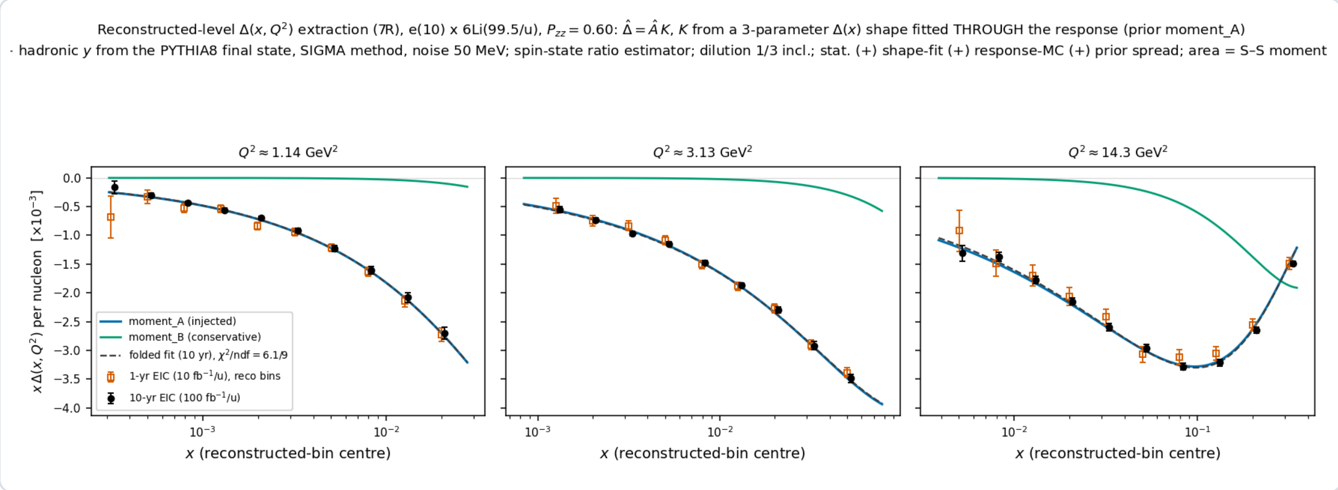
Sweet spot (reco bin)	25% Gaussian hadronic-y stand-in				Pythia 8 final state, Σ method, 50 MeV noise, calibrated					ΔÂ (10 yr) [10^{-5}]	ΔA, report 1 [10^{-4}]
	Purity	Eff.	D	Â ± ΔÂ (1 yr) [10^{-3}]	N (1 yr)	Purity	Eff.	D	Â ± ΔÂ vs reco-bin truth [10^{-3}]		
1: x = 0.028, Q² = 1.14	0.66	0.42	0.92	6.88 ± 0.12 (7.00)	1.6×10^8	0.56	0.31	1.00	7.38 ± 0.13 (7.61)	3.9	1.7
2: x = 0.011, Q² = 1.14	0.63	0.59	0.99	4.30 ± 0.09 (4.20)	3.0×10^8	0.59	0.56	1.06	4.61 ± 0.09 (4.50)	2.9	1.4
3: x = 0.071, Q² = 3.13	0.69	0.38	0.90	8.71 ± 0.16 (9.07)	8.8×10^7	0.64	0.32	0.95	9.78 ± 0.17 (9.56)	5.4	2.8
4: x = 0.141, Q² = 14.3	0.69	0.66	0.96	9.07 ± 0.29 (9.23)	2.9×10^7	0.75	0.71	0.98	9.78 ± 0.30 (9.41)	9.4	4.5

Table 4 — money plot 5R: the four sweet-spot super-bins in reconstructed (x, Q²) at e(10) × ⁶Li(99.5), P_{zz} = 0.6, model A, mixed method, spin-state ratio of the +0.6 / −1.2 states with the acceptance harmonic and the 10^{-3} luminosity offset on, statistical errors only. The reconstructed-bin truth is in parentheses. With the PYTHIA final state uncalibrated the purities would be 0.42 / 0.53 / 0.49 / 0.68 and D = 0.79 / 0.84 / 0.83 / 0.95; the errors are the same to within 7%. The last column is the single-state, truth-binned error of Report 1.

Three observations. The fit is unbiased: the eight pulls against the reconstructed-bin truth lie within 2 standard errors with the acceptance harmonic and the luminosity offset on, and the sin 2φ′ coefficients are consistent with zero. The errors are 0.59–0.69 of the single-state truth-level values of Report 1, because the 1.5× gain of the m = 0-rich state outweighs the reconstruction efficiency of 0.3–0.7 — presuming the source delivers that state at $|P_{zz}|$ = 1.2 with the same purity as the ±1 state at 0.6. And the real final state costs purity rather than statistics: the y ≈ 0.01 spots lose about 60% of their events to the reconstructed y ≥ 0.01 cut and a further third of the remainder to the Σ resolution, which is why more of the answer passes through the bin-centering factor there. Unfolded to Δ (money plot 7R), the best reconstructed bins reach statistical errors δΔ = 2.3 × 10^{-3} at Q² = 1.14 GeV², 1.1 × 10^{-3} at 3.13 GeV² and 0.3 × 10^{-3} at 14.3 GeV² in one year (1.7 / 1.7 / 6.9% of Δ; purities 0.44–0.56; 2.5 / 1.2 / 0.3 × 10^{-3} at 0.54–0.57 with the stand-in). With the bin-centering factor fitted through the response the full one-year bars at those bins are 4.2 / 2.6 / 7.0% and the ten-year ones 3.8 / 2.2 / 2.6%, the prior spread (3.7 / 2.1 / 1.3%) dominating at the two lower Q²; at Q² = 1.14 and 3.13 GeV² the ten-year data reject the flat-toy shape (χ²/ndf = 31/9 and 42/10 against 6/9 and 16/10 for the injected one), at 14.3 GeV² no member of the family is distinguishable.



Money plot 5R — inclusive gluonometry at the reconstructed level (e10 × ⁶Li99.5, P_{zz} = 0.6, model A, PYTHIA 8 final state with the Σ method and the calibrated hadronic scale, EMCal energy, track angles, $\varepsilon_{eID}(\eta)$, spin-state ratio of the +0.6 / −1.2 states with the acceptance harmonic and luminosity offset on). Left: the acceptance-free modulation T(φ′) − κ̂ per unit P_{zz} with the reconstructed-bin truth (blue) and the fit (vermillion). Right: the amplitude in reconstructed x bins along the Q² = 1.14 GeV² slice against the reconstructed-bin truth (blue ticks) and the model curves.



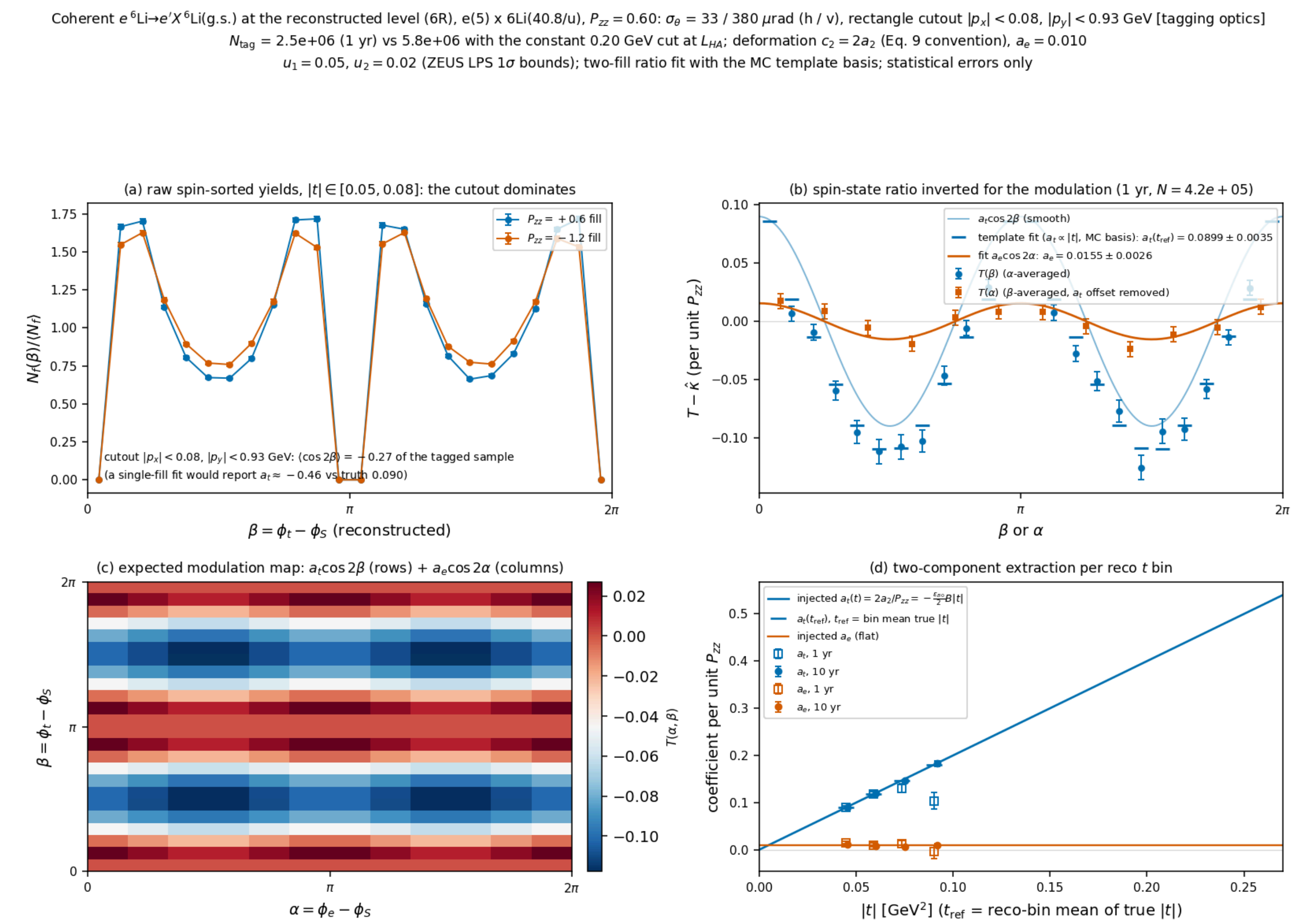
Money plot 7R — Δ(x,Q²) from reconstructed bins. xΔ points at the three sweet-spot Q² slices from the same pseudo-measurements, converted with the bin-centering factor fitted through the response (§4.4), so that the error bars carry the statistics, the shape fit, the response Monte Carlo and the prior spread; one-year (open) and ten-year (filled) draws are independent.

5.2 · Coherent: money plot 6R

CONFIGURATION	RECO |T| BIN [GEV²]	N_{TAG}, 1 YR	$A_T(T_{REF})$ INJECTED	FIT, 1 YR	ENSEMBLE MEAN, RATIO (20 EXP.)	ENSEMBLE MEAN, LIKELIHOOD (SAME 20 DRAWS)	FIT, 10 YR	A_E FIT, 1 YR (INJ. 0.010)	ENSEMBLE MEAN	A_E FIT, 10 YR
5 × 40.8, tag 41%, N_{tag} = 2.5×10⁶/yr	0.05–0.08	4.2×10⁵	0.090	0.0899 ± 0.0035	0.0899 (σ 0.0032)	0.0902 (σ 0.0033)	0.0892 ± 0.0011	0.0155 ± 0.0026	0.0092 (σ 0.0033)	0.0107 ± 0.0008
	0.08–0.12	1.8×10⁵	0.118	0.1184 ± 0.0048	0.1170 (σ 0.0038)	0.1176 (σ 0.0038)	0.1202 ± 0.0015	0.0103 ± 0.0040	0.0097 (σ 0.0037)	0.0084 ± 0.0012
	0.12–0.17	5.4×10⁴	0.147	0.131 ± 0.009	0.139 (σ 0.008)	0.146 (σ 0.009)	0.1465 ± 0.0027	0.012 ± 0.007	0.0101 (σ 0.0069)	0.0066 ± 0.0022
	0.17–0.25	1.3×10⁴	0.180	0.104 ± 0.018	0.114 (σ 0.023)	0.173 (σ 0.021)	0.183 ± 0.006	−0.004 ± 0.014	0.008 (σ 0.012)	0.0095 ± 0.0044
18 × 137.5, tag 32%, N_{tag} = 6.0×10⁶/yr	0.05–0.08	1.1×10⁶	0.100	0.1005 ± 0.0023	0.0998 (σ 0.0023)	0.0998 (σ 0.0023)	0.0992 ± 0.0007	0.0132 ± 0.0016	0.0096 (σ 0.0019)	0.0104 ± 0.0005
	0.08–0.12	4.1×10⁵	0.137	0.1333 ± 0.0033	0.1368 (σ 0.0041)	0.1372 (σ 0.0041)	0.1378 ± 0.0011	0.0148 ± 0.0026	0.0091 (σ 0.0028)	0.0108 ± 0.0008
	0.12–0.17	1.0×10⁵	0.179	0.1766 ± 0.0064	0.1776 (σ 0.0057)	0.1788 (σ 0.0055)	0.1789 ± 0.0020	0.0137 ± 0.0052	0.0097 (σ 0.0027)	0.0095 ± 0.0016
	0.17–0.25	1.9×10⁴	0.228	0.221 ± 0.015	0.207 (σ 0.016)	0.230 (σ 0.016)	0.226 ± 0.005	0.008 ± 0.012	0.0103 (σ 0.0102)	0.0130 ± 0.0037

Table 5 — money plot 6R: the two-azimuth template fit per reconstructed |t| bin at the tagging optics (coefficients per unit P_{zz}; ε_{B0} = −0.08, B = 50 GeV⁻²; u₁ = 0.05, u₂ = 0.02; luminosity at the tagging optics, 1/7.1 and 1/9.5 of the high-acceptance value; 6 × 10⁶ response recoils; `money_cos2phi_coherent_reco.py --optics tagging`). The "fit" columns are one pseudo-experiment; the ensemble columns are the mean and spread of twenty one-year pseudo-experiments on the same response, and say whether the estimator is biased. With more than 10⁵ tagged recoils per bin the ratio is not: the means agree with the injected values within their standard errors for a_t and a_e alike, and the single-experiment a_e of the first bin, 1.7 standard errors high, is the fluctuation the ensemble shows it to be. Below that count it is, and the two upper bins of 5 × 40.8 measure the defect at one year: 1451, 620, 192 and 48 counts per (α, β) cell give ratio means low by 0.0, −0.9, −5.4 and

−37%, while the same bins close at ten years. The likelihood column is the fix (§4.3) on the *same* twenty Poisson draws: 0.0902, 0.1176, 0.146 and 0.173 against 0.0898, 0.1181, 0.1473 and 0.1803 injected, and 0.0998, 0.1372, 0.1788 and 0.230 against 0.0997, 0.1372, 0.1791 and 0.2283 at 18 × 137.5. Two hundred experiments turn that into a measurement: the ratio's means are +0.1, −0.2, −4.0 and −34.3% from the injected values with pulls of the mean +0.4, −0.7, −9.0 and −42.1, the likelihood's +0.4, +0.3, +0.3 and +0.7% with pulls +2.0, +1.0, +0.5 and +0.9 (the first bin's residual is the response Monte Carlo's own floor at 6×10^6 recoils, not the estimator). The error bars become honest with it: the likelihood's spreads are 0.0031, 0.0048, 0.0091 and 0.0193 against quoted errors 0.0035, 0.0048, 0.0086 and 0.0187, where the ratio's quoted error in the sparsest bin is 13% below its own spread at 48 counts per cell and 62% above it at one count per cell. a_e is unbiased under both estimators, as the α-flat acceptance requires. Coarser binning only attenuates the defect: over the same two hundred experiments the sparsest bin's −34.3% at the default 12 × 24 (α, β) grid becomes −13.9% at 8 × 16 and −7.8% at 6 × 12, at a 17% cost on δa_e through the wider-bin dilutions (0.0139, 0.0148, 0.0162), and a 4 × 8 grid is rank-deficient at any count because ⟨cos 2α⟩ vanishes identically over four α bins; it is a cross-check, not the remedy. The ratio remains the default, so every number in the "fit" and ratio-ensemble columns is the published one. The sine columns are consistent with zero throughout and resolve a pot roll to 16–24 mrad per year in the three lower bins.



Money plot 6R — the coherent channel at the reconstructed level (5 × 40.8 at the tagging optics). **(a)** Raw spin-sorted yields versus the reconstructed recoil azimuth β in the lowest |t| bin: the cutout shape dominates both states identically. **(b)** The modulations T(β) and T(α) from the spin-state ratio, the template-fit prediction per β bin (ticks) and the smooth $a_t \cos 2\beta$ and $a_e \cos 2\alpha$ curves. **(c)** The expected modulation map in (α, β). **(d)** $a_t(t_{ref})$ and a_e per reconstructed |t| bin (1 yr open, 10 yr filled) against the injected curves.

6 · Systematic effects and the specifications they imply

Each effect was measured on the chain by holding the response fixed and varying one input with common random numbers (inclusive: `money_cos2phi_reco.py --syst-scan`; coherent: exact expected counts with the assumption changed on the fit side only). The response's own Monte-Carlo floor is 0.06–0.28% on $\hat{\Delta}$ at the four spots.

EFFECT	SIZE	ON THE RESULT
electron energy scale ±1%	unknown to the analysis	$\hat{\Delta}$ by 0.3–1.0% at the four spots (d ln x/d ln E′ = 2 − y with the Σ method)

hadronic energy scale, residual after calibration	+2%	$\hat{A}$ by +0.2 to +1.2% (0.1–0.6% per 1%)
hadronic resolution mis-modelled	generated at 30%, corrected with 25%	$\hat{\Delta}$ by 0.5–1.4%
η-dependent calorimetry; ε_{eID} η tilt of 0.05	—	≤ 0.1%; 0.02%
bin-centering model dependence	a wrong prior	4–9% at the spots with an assumed K, ≤ 1.8% (model B) / ≤ 4.6% (toy) with the shape fitted through the response
acceptance harmonic differing between the spin states	$\delta\varepsilon_2 = 10^{-3}$	fakes $\hat{A} = 5.6 \times 10^{-4}$ — 6–13% of the projected amplitudes, 2–6 one-year errors
relative luminosity between the states	10^{-3} on the ratio	0.03% (1/3 per unit ratio error, a constant in R)
P_{zz} scale	$\delta P_{zz}/P_{zz}$	1:1 on $\hat{\Delta}$, exactly (the ratio carries one power of the assumed scale against σ_P^2's two); a pure normalisation — every shape claim is immune
coherent: cutout differing between the states	horizontal half-width differing by 10^{-3} (10^{-2}) between the states	a_t by −0.8 / −0.5 / −0.1 / −0.3% (−7.6 / −4.5 / −3.6 / −2.9%) in the four |t| bins, a_e untouched — at the tagging cutout, whose horizontal edge sits inside a shallow part of the spectrum, a ~10^{-3} stability of the envelope is a per-mille effect (it was +19% under the earlier slot geometry)
coherent: u_2 assumed wrong	$\delta u_2 = 0.024$ (a ZEUS 1σ)	a_e by +0.0007 to +0.0009 (7–9%, 0.3 one-year errors) in the four |t| bins at 5 × 40.8 and +0.0008 to +0.0013 at 18 × 137.5, a_t untouched; a wrong u_1 moves only the nuisance a_m — first order through $a_t \delta u_2 \langle \cos 2\beta \rangle$. Measured in situ instead (next row) this falls to 0.5–6.3% of a_e
coherent: u_1, u_2 measured in situ	fit to the spin-averaged counts against the acceptance MC	δu_2 = 0.0028 / 0.0040 / 0.0070 / 0.0136 per |t| bin at 5 × 40.8 in one year (0.0016 / 0.0026 / 0.0049 / 0.0111 at 18 × 137.5), 1.8–15× tighter than the ZEUS band, the sparsest bin of either configuration the least improved; propagating into a_e as 0.00005 to 0.00063 across the eight bins, negligible in quadrature against the statistical error. The assumption moves rather than disappears: with the per-bin acceptance free, u is not identifiable at all (u_b multiplies ε_b identically), so an in-situ u necessarily rests on the acceptance shape the simulation supplies
coherent: relative luminosity	10^{-3}	0.1% on a_t, nothing on a_e

Table 6 — the systematic effects measured on the chain. Not modelled, and outside every error bar here: radiative corrections — §7 bounds the collinear-ISR migration at +0.5 to +1.2% of $\hat{\Delta}$ in the published generator window and ≤2.9% with the low-Q^2 feed-in opened up, but the tensor sector is uncalculated — backgrounds, and the F_2/R model (§7).

Two of the resulting specifications are this programme's to set, because no EIC document addresses a tensor-polarized beam: the tensor-polarization scale, $\delta P_{zz}/P_{zz} \le 5\%$ (3% has been reached on stored deuterons at the Nuclotron and on the HERMES storage cell, 8–10% is where solid-target work sits; Report 1 §7), propagating 1:1 into the magnitude of Δ and not at all into its shape; and the relative luminosity between the tensor states, ≤ 10^{-3}, which costs 0.03% and disappears with bunch-by-bunch alternation. The third is the acceptance-harmonic stability of 10^{-4} between the states, or the alternation that makes it exact. The others are asks with owners: $P_{zz} \ge 0.8$ from the source; a hadronic-method $\delta y/y \lesssim 0.3$ at $y \approx 0.01$ from ePIC full simulation, which turns on the calorimeter noise floor at Σ_h = 0.2–0.5 GeV; the lithium tagging optics and pots that follow it from C-AD and the far-forward group; charge identification in the pots at the 10^{-3} level for the $\alpha \leftrightarrow {}^6$Li separation; and a far-forward reconstruction that knows what lithium is, which is what this development supplies.

7 · What the chain does not model

Radiative corrections are not generated in the published response, and for the tensor sector they have never been calculated at all; what collinear initial-state radiation would cost is measured in the paragraph below, and it is the only part of the omission that can be bounded. Backgrounds are absent: no electron-identification purity or pair-symmetric background inclusively, no $\alpha + d$ breakup, vetoes or Z-identification in the coherent sample, and no beam-gas or pile-up. The coherent pseudo-experiment has no electron side and no x_P or Q^2 binning, and its template assumes the linear t shape it fits; the low-count bias of the bin-wise ratio estimator below a few hundred counts per (α, β) cell — already −4% at 192 and −34% at 48 (Table 5) — is no longer among these — the acceptance-profiled likelihood of §4.3 removes it (Table 5) and is available as an option, with the ratio kept as the default so that the numbers here are the ones the record carries. The fixed 0.05–0.25 GeV² window is a residual: it discards 67–74% of the tagged sample, and four bins added over 0.006–0.05 GeV² would halve the combined one-year δa_e, from 0.00205 to 0.00105, at the price of bins whose |t| is resolution-dominated. The structure functions are placeholders behind grid and R1998 backends. The track angular resolution and the calorimeter noise floor are stand-ins that the ePIC full simulation would replace, and the pot aperture was measured in a geometry that has since moved. The tagging optics is a proposal whose transport has not been computed. Each of these is listed in Table 2 or §6 with its leverage, and none changes the closure of §5, which is a statement about the chain given its inputs.

Collinear initial-state radiation is bounded rather than assumed. Every pseudo-event is given a photon of energy fraction z from the exponentiated leading-log electron structure function $D(z, Q^2) = (t/2)z^{t/2-1}S(t) - (t/4)(2-z)$, $t = (2\alpha/\pi)[\ln(Q^2/m_e^2) - 1]$, $S(t) = e^{(t/2)(3/4-\gamma_E)}/\Gamma(1+t/2)$ [21,22]; the hard vertex then sits at $\hat{s} = (1 - z)s$ while the analysis reconstructs with the nominal beam. The truncated exponentiation normalises to $1 + O(t^2)$, the residual being 7×10^{-4} at the t = 0.070 of the mid configuration's mean Q^2 = 4.4 GeV², where 27% of the selected rate radiates above $z = 10^{-4}$, $\langle z\rangle = 0.025$ and $\langle z\rangle = 0.092$ among those that do. Three of the four effects this could have are exactly zero. The covariant azimuth is invariant under $k \to (1-z)k$ for a massless target — $\cos\varphi'$ and $\sin\varphi'$ both acquire the factor $[2ac((1-z)a-b)]^{-1/2}$ with $a = P\cdot k$, $b = P\cdot k'$, $c = k\cdot k'$, and the arctangent divides it out — so radiation fakes neither a $\cos\varphi'$ nor a $\cos 2\varphi'$ modulation. The residual is 3.6×10^{-15} rad — double-precision roundoff — over a 2×10^4-event sample flat in (log x, log y) with z up to 0.9; over the 1.84×10^6 events of the response itself, where the physical ^{6}Li mass leaves the $O(\gamma^2)$ term of the projector, it reaches 2.6×10^{-2} rad and dilutes $\cos 2\varphi'$ by 9×10^{-8} rate-weighted. An unpolarized-lepton QED correction is common to the two fills by construction and cancels in the spin-state ratio of §4.3. And x is exact: the nominal s the analysis divides by exceeds the true $\hat{s}$ by $1/(1 - z)$, the electron-method Q^2 exceeds the hard Q^2 by the same $1/(1 - z)$, and y_Σ knows no beam energy [6], so $x = Q^2_e/(s\, y_\Sigma)$ is the hard x identically. What is left is the Q^2_e *label*, high by $1/(1 - z)$, and the migration it causes. Measured with common random numbers — the radiation draws from its own stream, so the two responses sit on the same pseudo-events — at 1600 pseudo-events per cell, the four sweet spots lose 0.015 to 0.044 in purity (0.653 → 0.638, 0.633 → 0.613, 0.679 → 0.659, 0.684 → 0.640) and 0.005 to 0.019 in efficiency, and Δ corrected with an ISR-free bin-centering is biased by +0.62 ± 0.03, +0.50 ± 0.02, +0.94 ± 0.03 and +1.22 ± 0.02%. A single response draw scatters by 4 to 14% of the bound — the seed-to-seed standard deviation is 0.05 to 0.09 percentage points and one seed can sit two of those from the mean — so those are means over eight response seeds and the errors are the standard errors of that mean, not a Monte Carlo floor. The generator window truncates the feed-in, since an event whose Q^2 lies below it cannot radiate into an analysis bin; opening it moves the worst spot to 1.87, 2.81, 2.26 and 2.34% at Q^2 = 0.35, 0.15, 0.05 and 0.02 GeV², so the bound saturates in a 1.8–2.8% band rather than at a single value, and ≤2.9% is what the ≤5% gate of plans/07 WP4 is read against. It passes by a factor 1.7. The number is not an artefact of the Gaussian y stand-in: run through the PYTHIA hadronic final state with the calibrated scale it is +0.38, +0.44, +0.46 and +0.88%. The method comparison behind the choice of estimator is sharp, and has to be read at a stated y because the electron-method ratios go as $(y + z)(1 - z)/y$. At z = 0.092 and the rate-weighted $\langle y\rangle = 0.189$ of the whole selected sample the observed-to-hard ratios (Q^2, y, x) are (1.102, 1.351, 0.740) for the electron method, (1.000, 1.000, 0.908) for Σ, (0.976, 0.908, 0.976) for Jacquet–Blondel, (1.214, 1.000, 1.102) for the double angle and (1.102, 1.000, 1.000) for the mixed pair the chain uses. At the sweet spots themselves, where y = 0.0102, 0.0256, 0.0111 and 0.0255, the same photon multiplies the electron-method y by 9.2, 4.2, 8.5 and 4.2 and collapses x_e to 0.11–0.24 of the hard x, quite apart from resolution; every other row is unchanged except Jacquet–

Blondel's Q^2, which rises to 0.998. Only the mixed row is 1.000 at every y. $Q^2_\Sigma = p_{T,e}^2/(1 - y_\Sigma)$ uses no beam energy either, so labelling the bin with it instead of Q^2_e would remove the migration entirely at the price of the Σ resolution on the label; that is a change to the chain and is not made here. Neither is the $E - p_z$ window every HERA analysis used against radiative events, which the chain also lacks: the visible sum is $2(1 - z)E_e$ and is already reconstructed, as $\Sigma_h + E'(1 - \cos\theta) = E'(1 - \cos\theta)/(1 - y_\Sigma)$, and requiring it within 15% of $2E_e$ keeps 87% of the non-radiative selected rate, 82% of the radiative one, and brings the bias to +0.23, +0.16, +0.22 and +0.18%, independent of the generator window. Outside all of this and unbounded: the tensor-sector radiative correction, which has never been calculated (plans/05 §5.5) and which no unpolarized study can stand in for; the polarized-lepton correction, irrelevant to an unpolarized beam; wide-angle real emission, which carries no logarithm and does not factorise into a beam-energy shift; final-state radiation, largely reabsorbed by the calorimeter cluster; and the elastic and quasi-elastic radiative tails, which the $W^2 \geq 10$ GeV² cut removes.

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Appendix A · Revision history and the 2026-08-28 review

DATE	REVISION

2026-08-28	Housekeeping from the integrated review: the reference list, which had skipped [15] and [16] since its first numbering, is renumbered so that the citations run 1–22 without a gap; §7 states the onset of the ratio's low-count bias at a few hundred counts per (α, β) cell rather than ~30, which is what Table 5 measures; Table 5's caption names the default 12 × 24 grid that the coarser binnings are compared against; and `harmonic_ratio_fit_2d` now returns `err_const`, so the ratio and the likelihood share the return shape the docstring advertises.
2026-08-28	**The coherent estimator's low-count bias closed, and the unpolarized harmonics measured in situ** (plans/08 A12, A3). The inversion of the bin-wise spin-state ratio is curved — concave for the flip plan, whose $\bar{P} = -0.3$ sets the sign — over the range the ratio explores at low counts, which put a negative per-bin offset of order 1/ν on $\hat{T}$ and — the acceptance being flat in α and modulated ×25 in β — landed it on a_t and the constant. The acceptance-profiled Poisson likelihood added alongside it (§4.3) is exactly the conditional multinomial given the bin totals, hence unbiased at any occupancy; Table 5 gains its column and §7 loses the corresponding caveat. The ratio remains the default, so no number previously published moved: the two agree to machine precision on exact counts and to 0.24% on the Asimov errors. (u_1, u_2) are now also fittable from the spin-averaged counts against the acceptance MC, which replaces the ZEUS 1σ band in Table 6 by a 1.8–15× tighter in-situ constraint and makes explicit that an in-situ u rests on the acceptance shape rather than on the data alone. The two-azimuth basis under a strongly anisotropic acceptance, listed as open since the aperture measurement, was closed by measurement: the design is near-orthogonal at the tagging optics (condition number 1.8–2.9, corr(a_e, a_t) < 0.005) and the anisotropy regime the item was written for became an empty sample when the ion energies were γ-matched.
2026-08-28	Radiative corrections bounded rather than assumed (plans/07 WP4, plans/08 D3, plans/02 step 1.4): `polligen/radiative.py` with the exponentiated leading-log photon spectrum, a default-off `RecoResponse(isr=…)` migration hook and `money_cos2phi_reco.py --isr`. §7 carries the result — collinear ISR fakes no cos φ′ or cos 2φ′, cancels in the spin-state ratio, leaves x exact and moves only the Q^2_e label — with the gate verdict (Δ biased by +0.5 to +1.2% in the published generator window, ≤ 2.9% once that window is opened below $Q^2 = 0.15$ GeV² and the low-Q^2 feed-in saturates, ≤ 0.25% behind an $E - p_z$ window, against a 5% gate) and the method comparison; a new row in Table 2. The bound is quoted as a mean over eight response seeds, one draw of the response scattering by about its own size; the method comparison is quoted with the y it is evaluated at, the electron-method rows being a strong function of it; and the azimuth residual is quoted per sample, the flat test sample and the response differing by two orders. 25 tests added; no published number moves, the hook being off by default.
2026-08-28	**Rewritten after a full review of the toolchain** (nine independent readings of the code against the questions: is each step right, does any analysis step read information an experiment cannot have, is every assumption stated; findings adversarially verified and recorded in `docs/code_review_2026-08-28.md`). What was wrong and is fixed: the hadronic-scale calibration was looked up at each pseudo-event's true (x, Q^2) cell — now a map in reconstructed (x_{mixed}, Q^2_e) bins built from the library's own reconstructed events, purities 0.56 / 0.59 / 0.64 / 0.75 against 0.52 / 0.59 / 0.59 / 0.76 before; the PYTHIA library carried PYTHIA's default $\hat{m} \geq 4$ GeV, which removed x < 16/s (39% of the selected rate) — regenerated with $\hat{m} \geq 0.5$ GeV; the estimator's bin variance vanished for a bin populated by one spin state (an infinite weight, and the "rank-deficient" |t| bins of the earlier Table 3) — now the expected-count form; the ratio inversion's error Jacobian dropped a second-order term; the hadron acceptance was applied in the head-on frame (+0.7–1.5% of Σ_h in the lab); the library's target-mass term was not carried onto the pseudo-events (2% of Σ_h at y = 0.01); the e + p / e + n merge was by count rather than cross section; the beam-energy spread had the same sign on Q^2_e and $1 - y_e$; the bins of the 5R panels were selected partly on the truth amplitude; the coherent pseudo-experiment ran on the legacy 73 μrad divergence with a 0.73 mrad "near-beam approach" — now the tagging optics of Report 1 with the Yellow Report divergences, at all three configurations, with a per-fill perturbation that acts on the binding cut; and the "residual template bias" on a_e was a 2σ fluctuation, shown by the ensemble of Table 5. The stand-in numbers of Table 4 did not move; the calibrated ones moved within their errors. Stale statements removed: the pre-correction sweet spots of the old Table 1, coherent numbers at 50 GeV/u, the 73/149 μrad footer, the 25% stand-in as the published hadronic y, the x → x/(1 − z) statement about radiation.
2026-08-24 – 27	First version and its revisions: the covariant azimuth and the head-on frame, the electron-method limit and the hadronic methods, the spin-state ratio, the coherent two-azimuth fit with the acceptance-weighted template basis and the sine null columns, the PYTHIA 8 final state through the hadron-side response, the Roman-Pot aperture measured in the ePIC geometry, the forward-folded bin-centering fit, the per-fill acceptance and the assumed nuisances as fit arguments, the γ-matched beam energies, the 1:1 propagation of the P_{zz} scale, and the consolidated assumptions and specifications. The numbers of those revisions that this one supersedes are recorded in plans/00 and plans/08.

Conventions: head-on frame with the ion along +z and the electron along −z; θ′ is the scattered-electron angle from the electron beam; per-nucleon (x, y, s) throughout; whole-nucleus four-momenta with the physical nuclear mass where a mass matters. Beam divergences are the Yellow Report Table 10.1 proton values with the γ-matched species step, and the tagging optics of Report 1 §6.1 where stated. Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.